

IEOR 4101: Probability, Statistics and Simulation

Lecture 7: Continuous Random Variables

Continuous Random Variables

- These are random variables taking values in *continuum*; e.g., in intervals
- E.g., cycle time of between cars coming off assembly lines in a Honda factory; amount of blood pressure drop when applying a new drug, etc

Probability Density Function

- **Definition.** The random variable X is *continuous* if there exists a *nonnegative* function $f(x)$ (defined on the whole real line) so that $f(x)$ integrates to 1, i.e.,

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

and for every real number a ,

$$\mathbb{P}(X \leq a) = \int_{-\infty}^a f(x)dx$$

- In this case, $f(x)$ is called the *probability density function* (pdf) of the random variable X
- The function $F(a) = P(X \leq a)$ is called the *cumulative distribution function* (cdf) of X

Some Discussion

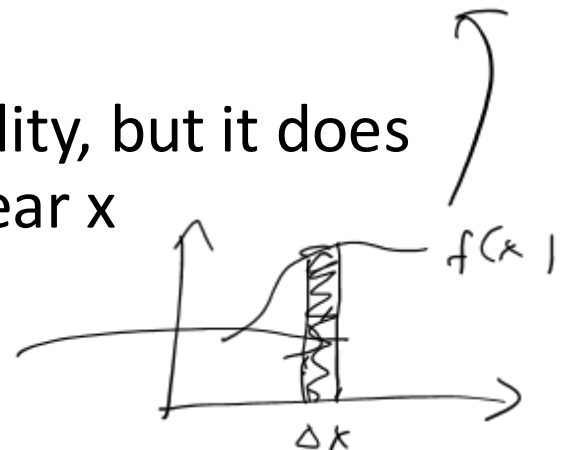
- **Remark 1.** $f(x)$ is the derivative of $F(a) = P(X \leq a)$
- **Remark 2.** Since F is differentiable, it is a continuous function. In particular, it has no “jumps” like the cdf of discrete random variables
- **Remark 3.** $f(x)$ is NOT $P(X = x)$ ($= 0$)
- **Remark 4.** One way to make sense of $f(x)$ is as follows

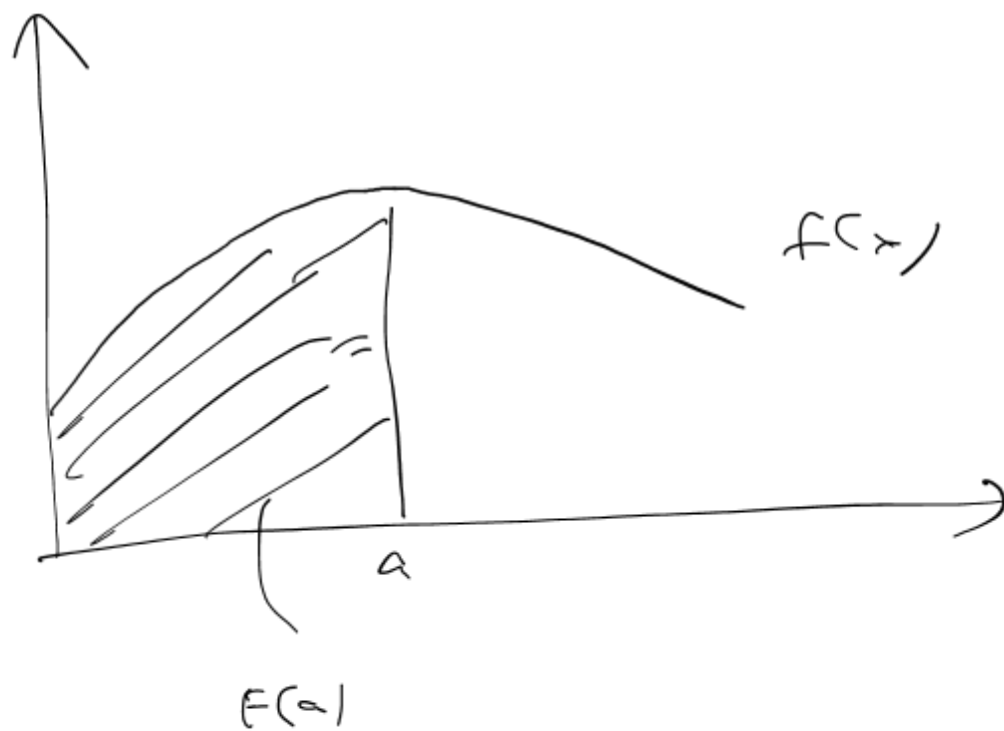
$$\mathbb{P}(a < X \leq b) = \mathbb{P}(X \leq b) - \mathbb{P}(X \leq a) = \int_a^b f(x) dx$$

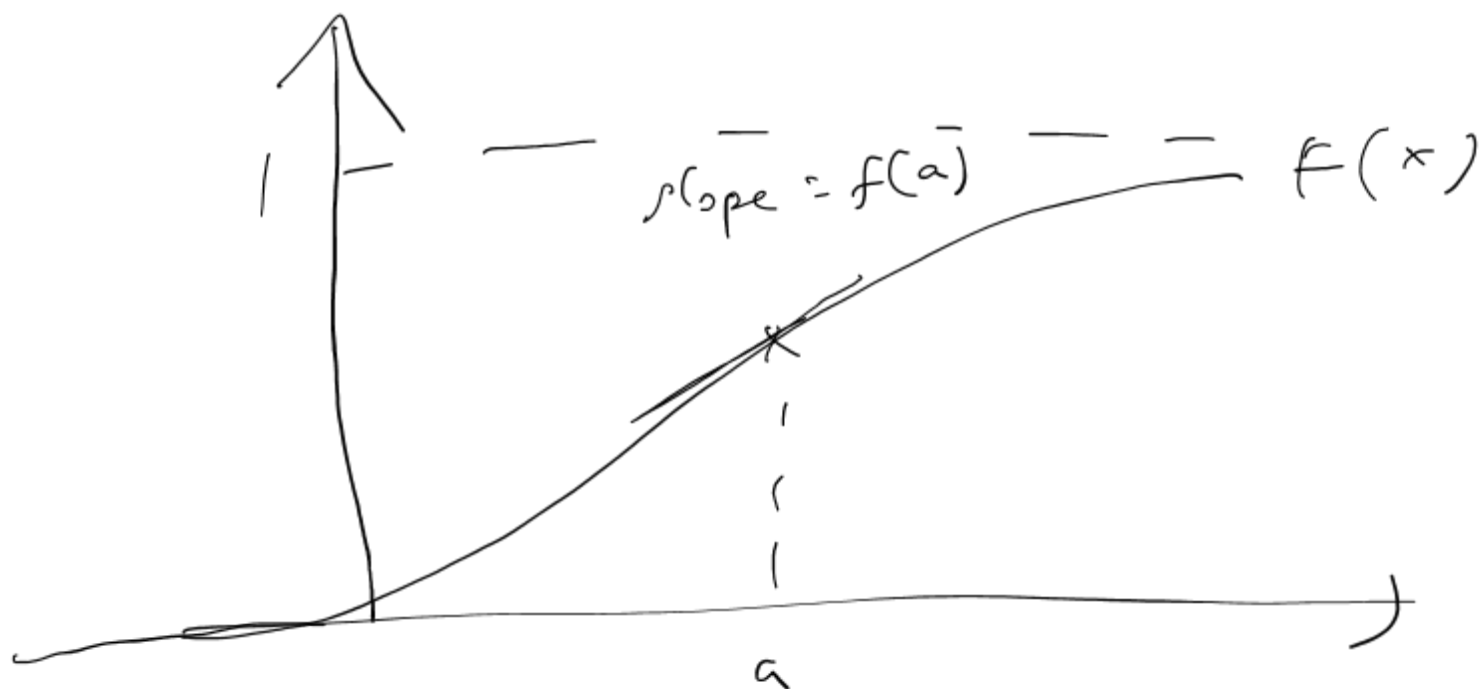
$$\mathbb{P}(x < X \leq x + \Delta x) = \int_x^{x+\Delta x} f(x) dx \approx f(x) \Delta x$$

for small Δx ; $f(x)$ itself is NOT a probability, but it does measure how likely it is to find a point near x

$$P(x < X \leq x + \Delta x) \approx f(x) \Delta x$$

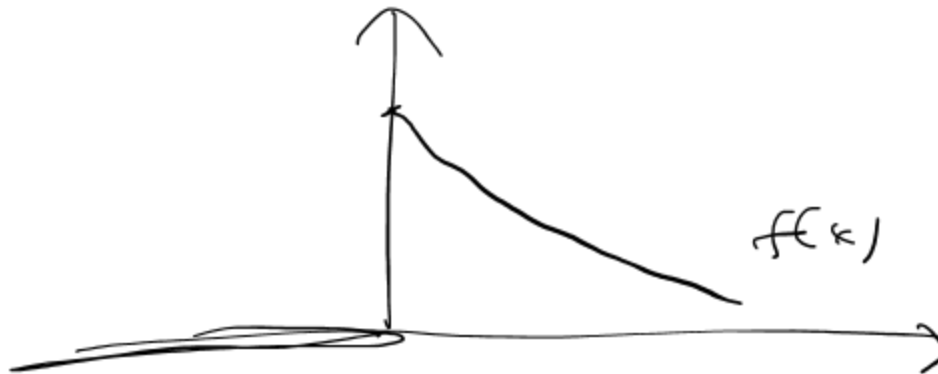






Probability Density Function

- Example.** The lifetime of an appliance can be modeled as a continuous random variable X with pdf $f(x)=c/(1+x)^3$ for $x > 0$ and $f(x) = 0$ otherwise. What is c ? Also find $P(X \leq 0.5)$, $P(0.5 < X \leq 1.5)$ and $P(0.5 < X \leq 1.5 \mid X > 0.5)$



We need
$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\Rightarrow \int_{-\infty}^0 0 dx + \int_0^{\infty} \frac{c}{(1+x)^3} dx = 1$$

$$\int_0^{\infty} \frac{c}{(1+x)^3} dx = \left. \frac{c}{2(1+x)^2} \right|_{\infty}^0$$

$$= \frac{c}{2}$$

$$\Rightarrow \frac{c}{2} = 1$$

$$\Rightarrow c = 2$$

$$P(X \leq 0.5) = \int_{-\infty}^{0.5} f(x) dx$$

$$= \int_0^{0.5} \frac{2}{(1+x)^3} dx$$

$$= \left. \frac{1}{(1+x)^2} \right|_0^{0.5}$$

$$= 1 - \frac{1}{1.5^2}$$

$$P(0.5 < x \leq 1.5) = \int_{0.5}^{1.5} f(x) dx$$

$$= \int_{0.5}^{1.5} \frac{2}{(1+x)^3} dx$$

$$= \left. \frac{1}{(1+x)^2} \right|_{0.5}^{1.5}$$

$$= \frac{1}{1.5^2} - \frac{1}{2.5^2}$$

$$P(0.5 < x \leq 1.5 \mid x > 0.5)$$

$$= \frac{P(0.5 < x \leq 1.5)}{P(x > 0.5)}$$

$$= \frac{\frac{1}{1.5^2} - \frac{1}{2.5^2}}{1 - \left(1 - \frac{1}{1.5^2}\right)}$$

Probability Density Function

- **Example.** The lifetime of an appliance can be modeled as a continuous random variable X with pdf $f(x)=c/(1+x)^3$ for $x > 0$ and $f(x) = 0$ otherwise. What is c ? Also find $P(X \leq 0.5)$, $P(0.5 < X \leq 1.5)$ and $P(0.5 < X \leq 1.5 \mid X > 0.5)$

- **Solution.** We need the pdf to integrate to 1, so

$$\begin{aligned}\int_{-\infty}^{\infty} f(x) dx &= \int_0^{\infty} \frac{c}{(1+x)^3} dx \\ &= c \left(-\frac{1}{2} \right) (1+x)^{-2} \Big|_0^{\infty} \\ &= 0 - \left(-\frac{c}{2} \cdot 1 \right) = \frac{c}{2} = 1\end{aligned}$$

- Thus $c = 2$

Probability Density Function

- **Example.** The lifetime of an appliance can be modeled as a continuous random variable X with pdf $f(x)=c/(1+x)^3$ for $x > 0$ and $f(x) = 0$ otherwise. What is c ? Also find $P(X \leq 0.5)$, $P(0.5 < X \leq 1.5)$ and $P(0.5 < X \leq 1.5 \mid X > 0.5)$
- **Solution continued.**

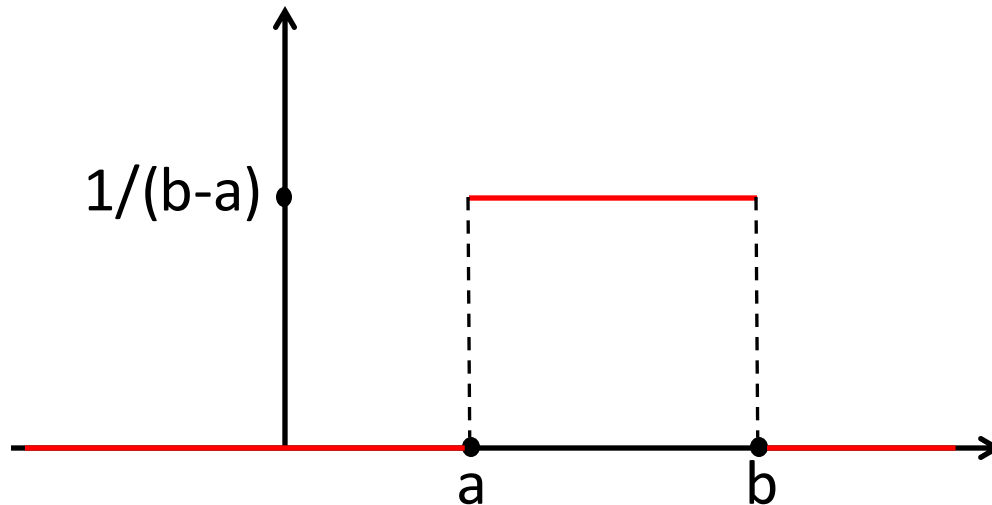
$$\begin{aligned}\mathbb{P}(X \leq 0.5) &= \int_0^{0.5} \frac{2}{(1+x)^3} dx \\ &= -(1+x)^{-2} \Big|_0^{0.5} \\ &= 1 - \left(\frac{1}{1.5}\right)^2 = \frac{5}{9}\end{aligned}$$

- Can also calculate $P(0.5 < X \leq 1.5) = 64/225$
- $P(0.5 < X \leq 1.5 \mid X > 0.5) = (64/225)/(1-5/9) = 16/25$

Uniform Random Variables

- A random variable X is uniformly distributed over $[a, b]$ if its probability density function (pdf) $f(x)$ is

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$



- Denote it by $\text{Unif}(a, b)$
- Used to model a “completely random” selection over the interval

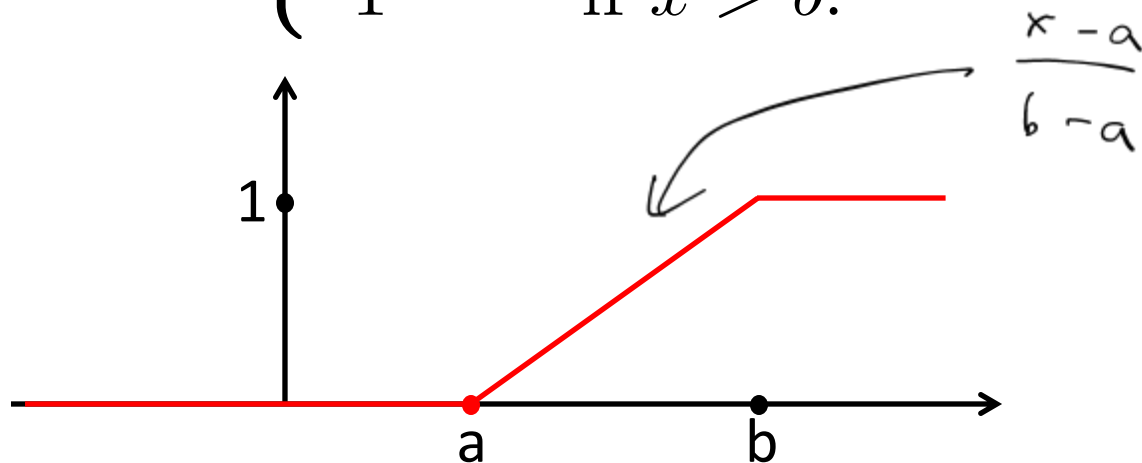
Uniform Random Variables

- Note that this is a valid pdf because

$$\int_a^b \frac{1}{b-a} dx = \frac{1}{b-a} \int_a^b dx = \frac{1}{b-a} \cdot (b-a) = 1$$

- The (cumulative) distribution function of $\text{Unif}(a, b)$ is

$$F(x) = \begin{cases} 0 & \text{if } x < a \\ \frac{x-a}{b-a} & \text{if } a \leq x \leq b \\ 1 & \text{if } x > b. \end{cases}$$



$$F(x) \quad \text{for} \quad a < x < b$$

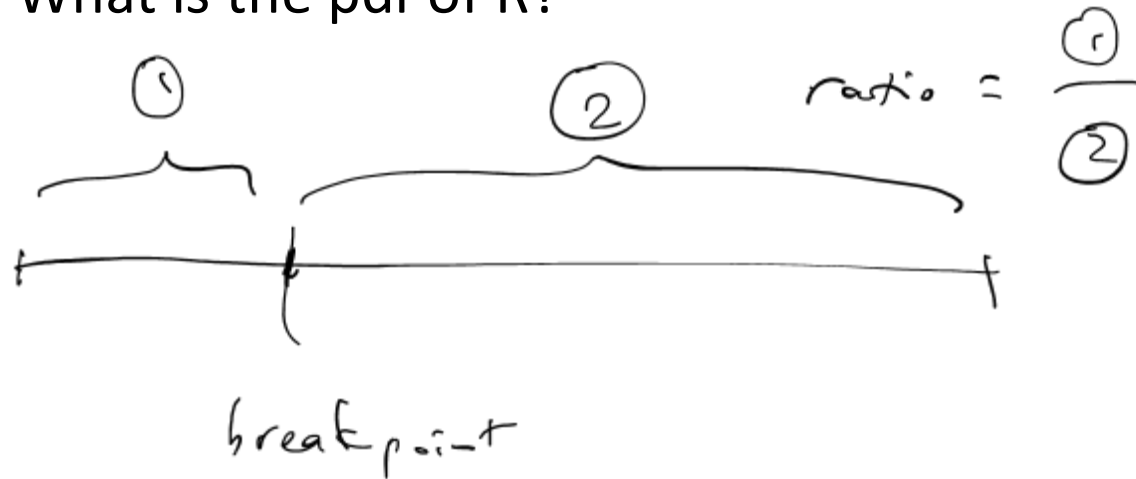
$$= \int_{-\infty}^x f(w) dw$$

$$= \int_a^x \frac{1}{b-a} dw$$

$$= \left. \frac{w}{b-a} \right|_a^x = \frac{x-a}{b-a}$$

Probability Density Function

- Example.** A chopstick of unit length is broken at random into two pieces. Let R be the ratio of the shorter piece to the longer piece. What is the pdf of R ?



We first find the cdf and then take derivative to get pdf.

cdf : $P(R \leq r)$

Suppose we break into two pieces,

u and $1-u$.

$$R = \min \left\{ \frac{u}{1-u}, \frac{1-u}{u} \right\}$$

$$P(R \leq r)$$

$$= P\left(\min\left\{\frac{u}{1-u}, \frac{1-u}{u}\right\} \leq r\right)$$

$$= P\left(\frac{u}{1-u} \leq r \text{ or } \frac{1-u}{u} \leq r\right)$$

$$= P(u \leq r(1-u) \text{ or } 1-u \leq ru)$$

$$= P\left(u \leq \frac{r}{1+r} \text{ or } u \geq \frac{1}{1+r}\right)$$

$$U \sim \text{unif}(0, 1)$$

$$P\left(U \leq \frac{r}{1+r} \text{ or } U \geq \frac{1}{1+r} \right)$$

$$= P\left(0 \leq U \leq \frac{r}{1+r} \right) + P\left(\frac{1}{1+r} \leq U \leq 1 \right)$$

$$= \int_0^{\frac{r}{1+r}} 1 \, dx + \int_{\frac{1}{1+r}}^1 1 \, dx$$

$$= \frac{r}{1+r} + \left(1 - \frac{1}{1+r} \right) = \frac{2r}{1+r}$$

Take derivative of $\frac{2r}{(1+r)}$ to get

$$\frac{d}{dr} \left(\frac{2r}{1+r} \right) = \frac{d}{dr} \left(2 \left(1 - \frac{1}{1+r} \right) \right)$$

$$= \frac{2}{(1+r)^2}$$

So pdf is $\frac{2}{(1+r)^2}$ for $0 < r < 1$ and 0 otherwise.

Probability Density Function

- **Example.** A chopstick of unit length is broken at random into two pieces. Let R be the ratio of the shorter piece to the longer piece. What is the pdf of R ?
- **Solution.** Consider the outcome u as the point at which the chopstick is broken from 0. $R(u) = \min\{u/(1-u), (1-u)/u\}$
- The event $\{u: R(u) \leq r\}$ means either $u/(1-u) \leq r$ or $(1-u)/u \leq r$
- This is equivalent to $\{1/(1+r) \leq u < 1\} \cup \{0 < u \leq r/(1+r)\}$
- $P(R \leq r) = P(1/(1+r) \leq u < 1 \text{ or } 0 < u \leq r/(1+r)) = 2r/(1+r)$
- Derivative of $2r/(1+r)$ w.r.t r is $2/(1+r)^2$
- $f(x) = 2/(1+x)^2$ for $0 < x < 1$, and equal 0 otherwise

Joint Probability Density Function

- Similar to the discrete case, we can define joint distributions for continuous random variables
- **Definition.** Continuous random variables X and Y have a joint pdf $f(x, y)$ if $f(x, y) \geq 0$ for all x and y ,

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1,$$

and for all $-\infty < a, b < \infty$,

$$\mathbb{P}(X \leq a, Y \leq b) = \int_{-\infty}^b \int_{-\infty}^a f(x, y) dx dy$$

Discussion

- Similar to the one-dimensional case, the following interpretation applies:

$$P(X \in (a, a + \Delta a), Y \in (b, b + \Delta b)) \approx f(a, b) \Delta a \Delta b$$

where Δa and Δb are small

- The pdf $f(x, y)$ can be derived from the cdf $P(X \leq x, Y \leq y)$ as follows:

$$f(x, y) = \frac{\partial^2}{\partial x \partial y} \mathbb{P}(X \leq x, Y \leq y)$$

Jointly Continuous RV

- Example.** An electronic system has two crucial components. The system fails if either component fails. The joint density function of the lifetimes (in units of years) X and Y of the two components is $f(x, y) = (1+x+y)/12$ for $0 < x, y < 2$ and $f(x, y) = 0$ otherwise. What is the probability that the system will fail in the first year?

$$P(\text{fail in 1st yr})$$

$$= P(X \leq 1 \text{ or } Y \leq 1)$$

$$= P(X \leq 1) + P(Y \leq 1) - P(X \leq 1, Y \leq 1)$$

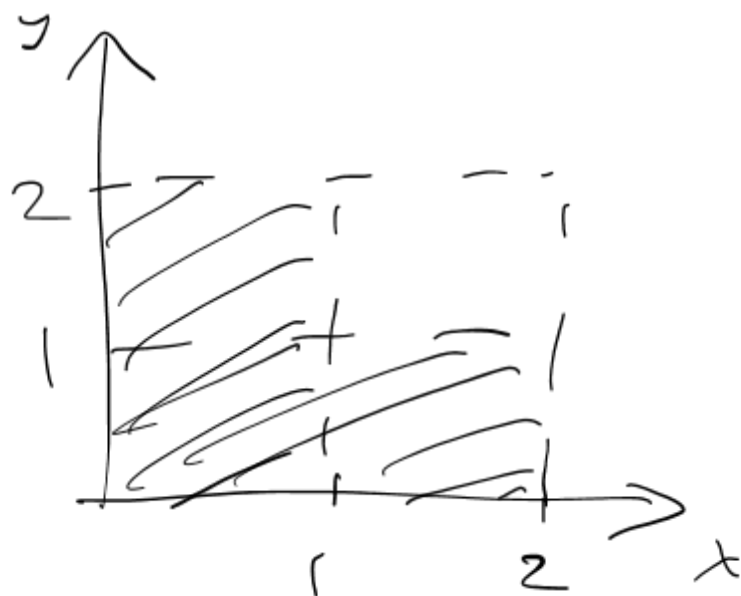
$$= \dots$$

$$P(X \leq 1 \text{ or } Y \leq 1)$$

$$= 1 - P(X > 1 \text{ and } Y > 1)$$

$$= 1 - \int_1^2 \int_1^2 f(x, y) dx dy$$

$$= 1 - \int_1^2 \int_1^2 \frac{1+x+y}{12} dx dy$$



$$\int_1^2 \int_1^2 \frac{1+x+y}{12} dx dy$$

$$= \int_1^2 \left. \frac{1+y}{12} x + \frac{x^2}{24} \right|_1^2 dy$$

$$= \int_1^2 \frac{1+y}{12} + \frac{3}{24} dy$$

$$= \int_1^2 \left(\left(\frac{1}{12} + \frac{1}{8} \right) + \frac{y}{12} \right) dy$$

$$= \left(\frac{1}{12} + \frac{1}{8} \right) y + \frac{y^2}{24} \Big|_1^2$$

$$= \left(\frac{1}{12} + \frac{1}{8} \right) + \frac{3}{24}$$

$$= \dots$$

Jointly Continuous RV

- **Example.** An electronic system has two crucial components. The system fails if either component fails. The joint density function of the lifetimes (in units of years) X and Y of the two components is $f(x, y) = (1+x+y)/12$ for $0 < x, y < 2$ and $f(x, y) = 0$ otherwise. What is the probability that the system will fail in the first year?
- **Solution.** We are interested in $P(X \leq 1 \text{ or } Y \leq 1) = 1 - P(X > 1 \text{ and } Y > 1)$

$$\begin{aligned} &P(X > 1, Y > 1) \\ &= \int_1^2 \int_1^2 \frac{1}{12} (1 + x + y) \, dx dy = \frac{1}{12} \int_1^2 \left(1 + x + \frac{3}{2} \right) dx = \frac{1}{3} \end{aligned}$$

- So $P(X \leq 1 \text{ or } Y \leq 1) = 2/3$

Independent Continuous RV

- **Definition (Marginal pdf).** Let X and Y be continuous random variables with joint pdf $f(x, y)$. The marginal pdf f_X of X is given by

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

- and the marginal pdf f_Y of Y is given by

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

- Can show that (similar for Y)

$$P(X \leq x) = \int_{-\infty}^x f_X(u) du$$

- **Definition (Independence).** Continuous rv X and Y are independent if for any x and y ,

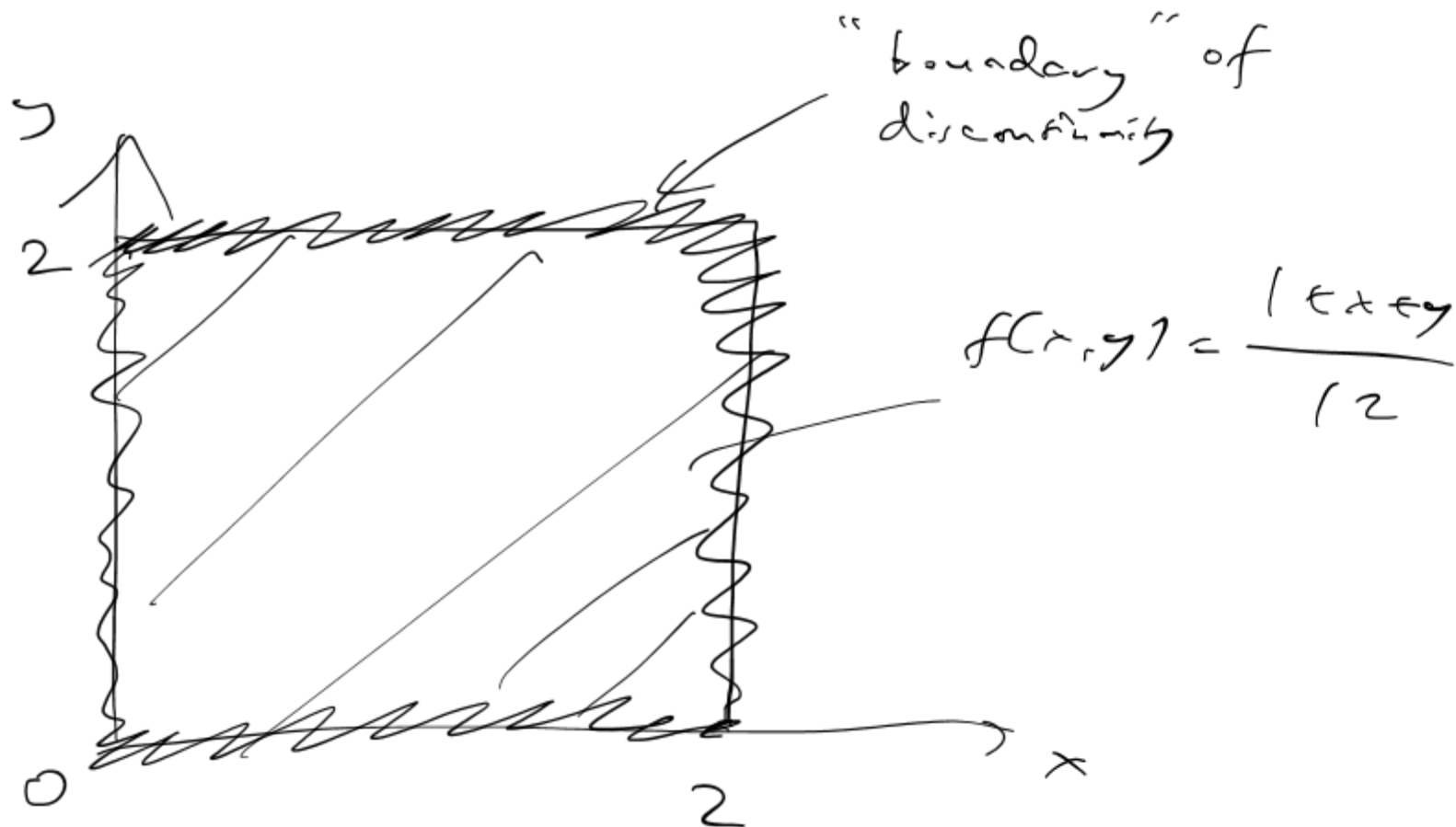
$$P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y)$$

or equivalently, for any x and y not at the “boundaries” of discontinuity

$$f(x, y) = f_X(x)f_Y(y)$$

← can also be used in discrete case

← similar to discrete case



Independent Continuous RV

- **Example.** in our previous example where X and Y are distributed according to $f(x, y) = (1+x+y)/12$ for $0 < x, y < 2$ and $f(x, y) = 0$ otherwise, are X and Y independent?
- **Example.** You await at a bus stop. You can take two different buses to get to your destination. The arrival times X and Y of the two buses has a joint pdf $f(x, y) = 6e^{-(2x+3y)}$, for $x > 0$ and $y > 0$, and 0 otherwise. Are X and Y independent? What is the pdf of $\min\{X, Y\}$?

Joint pdf:

$$f(x, y) = \frac{1+x+y}{12} \quad \text{for} \quad \begin{matrix} 0 < x < 2 \\ 0 < y < 2 \end{matrix}$$

Marginal pdf:

$$f_x(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

$$= \int_0^2 \frac{1+x+y}{12} dy = \left. \frac{1+x}{12} y + \frac{y^2}{24} \right|_0^2$$

$$= \frac{1+x}{6} + \frac{1}{6} = \frac{2+x}{6}$$

$$f_X(x) = \frac{2+x}{6} \quad \text{for } 0 < x < 2$$

$$= 0 \quad \text{otherwise}$$

$$f_Y(y) = \frac{2+y}{6} \quad \text{for } 0 < y < 2$$

$$0 \quad \text{otherwise}$$

$$\frac{1+x+y}{12} \stackrel{?}{=} \frac{2+x}{6} \cdot \frac{2+y}{6}$$

Pick $x=1, y=1$

$$\frac{1+1+1}{12} \stackrel{?}{=} \frac{2+1}{6} \cdot \frac{2+1}{6}$$

Pick $x = 1, y = \frac{1}{2}$

$$\frac{1 + 1 + \frac{1}{2}}{12} = \frac{2 + 1}{6} \cdot \frac{2 + \frac{1}{2}}{6}$$

Pick $x = 1, y = \frac{3}{2}$

$$\frac{1 + 1 + \frac{3}{2}}{12} = \frac{2 + 1}{6} \cdot \frac{2 + \frac{3}{2}}{6}$$

Pick $x = \frac{3}{2}$, $y = \frac{1}{2}$

$$\frac{1 + \frac{3}{2} + \frac{1}{2}}{12} \neq \frac{2 + \frac{3}{2}}{6} \cdot \frac{2 + \frac{1}{2}}{6}$$

Hence X and Y are not independent

Suppose we answer by saying:

Pick $x = 2, y = 2$

$$\frac{1+2+2}{12} \neq \frac{2+2}{6} \cdot \frac{2+2}{6}$$

Hence X and Y are not independent.

$$f(x, y) = 6e^{-(2x+3y)} \quad \text{for } x > 0, y > 0$$

= 0 otherwise.

$$f_x(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_0^{\infty} 6e^{-(2x+3y)} dy$$

$$= 6e^{-2x} \int_0^{\infty} e^{-3y} dy$$

$$= 2e^{-2x}$$

$$f_x(x) = \begin{cases} 2e^{-2x} & \text{for } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

$$f_y(y) = \begin{cases} 3e^{-3y} & \text{for } y > 0 \\ 0 & \text{otherwise} \end{cases}$$

$$6e^{-(2x+3y)} = 2e^{-2x} \cdot 3e^{-3y}$$

for any $x, y > 0$.

Hence X and Y are independent.

Alternative approach:

Note that joint pdf can be written as

(a function that only depends on x)

$\times$ (a function that only depends on y).

$\Rightarrow X$ and Y must be independent.

$$6e^{-(2x+3y)} = \underbrace{6e^{-2x}}_{\text{only depends, on } x} \cdot \underbrace{e^{-3y}}_{\text{only depends on } y}.$$

cdf:

$$P(\min\{X, Y\} \leq z)$$

$$= P(X \leq z \text{ or } Y \leq z)$$

$$= 1 - P(X > z \text{ and } Y > z)$$

$$= 1 - P(X > z) P(Y > z)$$

because X and
 Y are independent

$$= 1 - \int_z^{\infty} f_X(x) dx \int_z^{\infty} f_Y(y) dy$$

$$= 1 - \int_z^{\infty} 2e^{-2x} dx \int_z^{\infty} 3e^{-3y} dy$$

$$= 1 - e^{-2z} \cdot e^{-3z}$$

$$= 1 - e^{-5z}$$

Take derivative to get

$$f(z) = \begin{cases} 5e^{-5z} & \text{for } z > 0 \\ 0 & \text{otherwise} \end{cases}$$

as the pdf of $\min\{X, Y\}$

Conditional Probabilities

- For a pair of joint discrete r.v.'s X and Y , with pmf $p(x, y)$, the conditional pmf of X given $Y=y$ is

$$p_{X|Y}(x|y) = P(X = x|Y = y) = \frac{p(x, y)}{p_Y(y)}$$

- The conditional cdf of X given $Y=y$ is

$$F_{X|Y}(x|y) = P(X \leq x|Y = y) = \sum_{a \leq x} p_{X|Y}(a|y) = \frac{\sum_{a \leq x} p(a, y)}{p_Y(y)}$$

Conditional Probabilities

- For a pair of joint continuous r.v.'s X and Y , with pdf $f(x, y)$, the conditional pdf of X given $Y=y$ is

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}$$

- The conditional cdf of X given $Y=y$ is

$$F_{X|Y}(x|y) = \int_{-\infty}^x f_{X|Y}(u|y) du = \frac{\int_{-\infty}^x f(u, y) du}{f_Y(y)}$$

from the view of conditional prob,
it does not make sense because $f_Y(y)$ is
not a prob, and also $P(Y=y)=0$ since Y is continuous.

Conditional Probabilities

- **Example.** in our previous example where X and Y are distributed according to $f(x, y) = (1+x+y)/12$ for $0 < x, y < 2$ and $f(x, y) = 0$ otherwise, what are the conditional pdf and cdf of X given $Y=1$?

$$f_{X|Y}(x|1) = \frac{f(x, 1)}{f_Y(1)} = \frac{\frac{1+x+1}{12}}{\frac{2+1}{6}}$$

$$= \frac{1}{3} + \frac{x}{6} \quad \text{for } 0 < x < 2$$

and 0 otherwise

$$F_{X|Y}(x|1) = \int_0^x \left(\frac{1}{3} + \frac{u}{6} \right) du$$

$$= \left[\frac{1}{3}u + \frac{u^2}{12} \right]_0^x$$

$$= \frac{1}{3}x + \frac{x^2}{12} \quad \text{for } 0 < x \leq 2$$

and $= 0$ for $x \leq 0$, $= 1$ for $x > 2$.

Expectation of Continuous RV

- Recall the definition of expectation for discrete rv
- Suppose X takes values $x_1, x_2, \dots$ and $P(X = x_i) = p_i$
- Then $E[X] = p_1x_1 + p_2x_2 + \dots$
- **Definition.** The *expectation* of a continuous random variable X with pdf $f(x)$ is defined by

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f(x) dx$$

- **Remark.** $f(x)dx$ serves as the continuous analog of p_i in the discrete case

Expectation of Continuous RV

- **Example.** A chopstick of unit length is broken at random into two pieces. Let R be the ratio of the shorter piece to the longer piece. What is $E[R]$?
- **Solution.** We know that $P(R \leq r) = 2r/(1+r)$ for $0 < r < 1$
- $f_R(r) = 2/(1+r)^2$, $0 < r < 1$; $f_R(r) = 0$ otherwise
- Then,

$$\begin{aligned}\mathbb{E}[R] &= \int_{-\infty}^{\infty} r f_R(r) dr = \int_0^1 r \cdot \frac{2}{(1+r)^2} dr \\ &= 2 \int_0^1 \frac{1}{1+r} - \frac{1}{(1+r)^2} dr \\ &= 2 \log(r+1) \Big|_0^1 - 2 \left(-\frac{1}{r+1} \right) \Big|_0^1 = 2 \log 2 - 1\end{aligned}$$

Expectation of Functions and Variance

- Similarly to the discrete case, for a continuous random variable X with pdf $f(x)$, suppose we are interested in the expectation of $g(X)$ where $g(x)$ is a function of x . Then

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x)f(x)dx$$

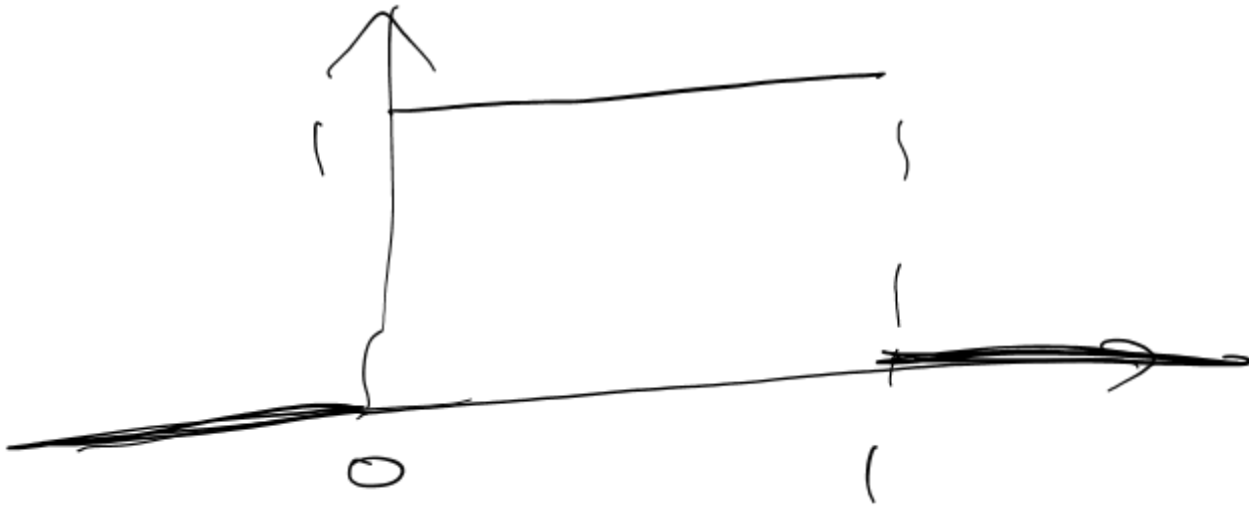
- **Definition.** The variance $\text{Var}(X)$ of a continuous r.v. X is defined to be $\text{Var}(X) = E[(X-\mu)^2]$ where $\mu = E[X]$
- Can show $\text{Var}(X) = E[X^2] - E^2[X]$

Expectation of Functions and Variance

- **Example.** U is a number that is uniformly distributed on the interval $(0, 1)$. What is the variance of U ?

pdf:

$$f(x) = \begin{cases} 1 & \text{if } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$



$$\mu = E[x] = \int_{-\infty}^{\infty} x f(x) dx = \int_0^1 x dx = \frac{x^2}{2} \Big|_0^1 = \frac{1}{2}$$

$$\text{Var}(x) = E(x - \mu)^2$$

$$\text{or } = E[x^2] - (E[x])^2$$

$$= \int_0^1 x^2 dx - \left(\frac{1}{2}\right)^2$$

$$= \frac{x^3}{3} \Big|_0^1 - \left(\frac{1}{2}\right)^2$$

$$= \frac{1}{12}$$

Expectation of Functions and Variance

- **Example.** U is a number that is uniformly distributed on the interval $(0, 1)$. What is the variance of U ?

- **Solution.** The pdf of U is $f(u) = 1$ for $0 < u < 1$ and 0 otherwise. The expectation is

$$\mathbb{E}[U] = \int_0^1 u du = \frac{1}{2} u^2 \Big|_0^1 = \frac{1}{2}$$

- So $\mu = 0.5$. Now

$$\begin{aligned} \text{Var}(U) &= \mathbb{E}[(U - 0.5)^2] = \int_0^1 (u - 0.5)^2 du \\ &= \frac{1}{3} (u - 0.5)^3 \Big|_0^1 = \frac{1}{6} - \left(\frac{1}{6} \right) = \frac{1}{12} \end{aligned}$$

Properties of Expectation and Variance

- Similarly, the covariance between two continuous r.v.'s X and Y is

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

- All properties of expectation, variance and covariance for discrete r.v.'s hold also for continuous r.v.'s. For example,
- $E[X+Y] = E[X] + E[Y]$
- $E[aX+b] = aE[X] + b$
- $\text{Var}(aX+b) = a^2\text{Var}(X)$
- $\text{Cov}(X, Y) = E[(X-\mu_X)(Y-\mu_Y)] = E[XY] - E[X]E[Y]$
- $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$
- If X and Y are independent, then $E[XY] = E[X]E[Y]$, $\text{Cov}(X, Y) = 0$, and $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$